

RED SEAL EXAM PREP

456A Welder

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

Pass mark: 70% · Aligned to the published occupational standard topic weightings
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Mock Exam — answer all questions, then check the key

Q1. What is the function of anti-spatter spray in GMAW?

- A) To reduce oxidation of the weld pool
- B) To lubricate the wire liner
- C) To improve shielding gas coverage
- D) To keep spatter from sticking to the nozzle

Q2. Before entering a confined space for welding, what is the FIRST action?

- A) Put on the welding helmet and gloves
- B) Turn on the ventilation fans
- C) Test the air with a gas monitor
- D) Notify a co-worker standing by outside

Q3. What does "stringer bead" mean in multi-pass welding?

- A) A bead deposited in the vertical-up position
- B) A narrow bead deposited with no side-to-side motion
- C) A wide weave bead deposited side to side
- D) The final cap pass over a multi-pass weld

Q4. What type of tungsten tip preparation is used for GTAW on steel with DCEN?

- A) A split (notched) tip
- B) A sharp tapered point
- C) A rounded ball tip
- D) A flat-ground tip

Q5. Which filler metal is used for GTAW of chrome-moly (Cr-Mo) steel such as P91?

- A) ER308L
- B) ER70S-2
- C) ER90S-B9
- D) ER4043

Q6. What is the purpose of the high-frequency (HF) start in GTAW?

- A) To generate the shielding gas ionization
- B) To start the arc without touching the workpiece
- C) To increase amperage for thicker material
- D) To provide the cleaning action for aluminum

Q7. Which GMAW transfer mode requires the highest arc voltage to achieve?

- A) Short circuit transfer
- B) Pulsed spray transfer
- C) Globular transfer
- D) Spray transfer

Q8. What is the purpose of preheating steel before SMAW?

- A) To improve slag fluidity for easier removal
- B) To increase arc voltage
- C) To slow the cooling rate of the joint
- D) To increase the deposition rate

Q9. What is plasma arc cutting (PAC) best suited for compared to OFC?

- A) Cutting materials that do not conduct electricity
- B) Cutting galvanized steel outdoors
- C) Cutting stainless steel and aluminum
- D) Cutting thick carbon steel over 100 mm

Q10. Excessive spatter in FCAW-G is most likely caused by:

- A) Wire stickout too short
- B) Shielding gas flow rate too high
- C) Arc voltage set too low
- D) Base metal temperature too high

Q11. ER70S-6 wire has higher silicon and manganese content than ER70S-3. Why?

- A) To allow welding with 100% CO₂ shielding gas only
- B) To enable AC power source compatibility
- C) To add deoxidizers for welding slightly rusty surfaces
- D) To increase tensile strength above 80 ksi

Q12. In the FCAW electrode classification E71T-1, what do the "T" and the trailing "-1" designate?

- A) T = travel direction; -1 = flat position only
- B) T = tubular electrode; -1 = single-pass use only
- C) T = titanium flux coating; -1 = wire diameter
- D) T = tubular electrode; -1 = CO₂ or Ar/CO₂, DCEP

Q13. What is the Lower Explosive Limit (LEL) of acetylene in air?

- A) 1.0%
- B) 9.5%
- C) 2.5%
- D) 4.3%

Q14. What is the Heat Affected Zone (HAZ) in welding?

- A) The area heated by the torch before welding starts
- B) The area of the weld covered by the shielding gas
- C) Base metal altered by welding heat but not melted
- D) The weld metal itself, meaning the fusion zone

Q15. Low-hydrogen electrodes (E7018) must be stored in:

- A) A heated rod oven at 120–150°C (250–300°F)
- B) A freezer to prevent oxidation
- C) Any dry location away from direct sunlight
- D) A sealed plastic bag at room temperature

Q16. During GMAW the arc pops and stutters, the wire occasionally burns back to the contact tip, and the bead comes out irregular. What is the most likely cause?

- A) Work angle too steep for the joint
- B) Worn drive rolls or a kinked liner
- C) Travel speed too slow for the amperage
- D) Shielding gas flow rate set too low

Q17. Carbon equivalent (CE) formula is used to determine:

- A) The post-weld heat treatment temperature
- B) The tensile strength of the completed weld
- C) The hardenability and preheat requirement
- D) The carbon content of the weld metal

Q18. What is globular transfer in GMAW and when does it occur?

- A) Transfer of large droplets at intermediate voltage — high spatter
- B) Transfer of very fine droplets at high current — preferred for production welding
- C) Transfer where the wire dips into the molten pool at low voltage
- D) Transfer mode unique to pulsed GMAW at low background current

Q19. After chipping the slag off a flux-cored fillet, the welder finds narrow elongated grooves running along the crown of the bead. What are they?

- A) Undercut, where the arc melted the toe faster than the puddle refilled it
- B) Arc strikes, where the wire touched the plate outside the joint on restart
- C) Worm tracks, from gas trapped between the weld metal and freezing slag
- D) Slag inclusions, where flux rolled ahead of the arc and was welded over

Q20. In oxyfuel cutting (OFC), what initiates the cutting action?

- A) A high-pressure jet of pure oxygen
- B) An electric arc combined with oxygen
- C) The heat from the preheat flames alone
- D) The acetylene flame melting the metal

Q21. Why must oxygen hoses and fittings be kept free of oil and grease?

- A) Grease reduces gas flow rate
- B) Oil causes corrosion of brass fittings
- C) They can ignite spontaneously in pure oxygen
- D) Oil contaminates the flame and changes its characteristics

Q22. A 456A welder is welding mild steel at a fixed indoor workstation. Under CSA W117.2 and provincial occupational health and safety regulation, what determines whether the ventilation is adequate?

- A) 1,000 cfm of general dilution air per welder whenever the shop ceiling is under 5 m
- B) Nothing below 100 A — ventilation applies only to high-current welding on coated metal
- C) Fume kept below the occupational exposure limits, with local exhaust at the source
- D) 2,000 cfm of general dilution air per welder, the fixed minimum set by Canadian regulation

Q23. What does "DCEP" mean in SMAW?

- A) Direct Current Electrode Positive — electrode is negative, work is positive
- B) Direct Current Electrode Polarity — a setting found only on AC machines
- C) Direct Current Electrode Positive — electrode is positive, work is negative
- D) Dual Current Electrode Positive — a machine setting allowing both AC and DC

Q24. What does "walking the cup" technique mean in GTAW?

- A) Bracing the nozzle cup against the base metal and rocking it to move the torch steadily
- B) Oscillating the filler rod in and out of the leading edge of the puddle to control bead width
- C) Moving the torch in a wide circular pattern so the heat spreads evenly across the joint and the puddle stays fluid
- D) Whipping the torch back and forth to let the puddle cool between advances

Q25. What is the purpose of the liner in a GMAW torch?

- A) To regulate shielding gas flow
- B) To insulate the torch from the work cable
- C) To guide the wire to the contact tip
- D) To cool the contact tip

Q26. E6013 is preferred for welding thin sheet metal because:

- A) It provides maximum penetration into thin material
- B) It produces a hard slag that prevents porosity
- C) It has a soft arc, easy slag removal, and works on AC or DC
- D) It generates the highest amperage range

Q27. What causes "sugaring" (oxidation/discolouration) on the back side of a stainless steel GTAW root pass?

- A) Lack of back purge on the root side
- B) Excessive heat input from high amperage
- C) Insufficient post-flow gas time
- D) Using the wrong filler wire for the base metal

Q28. Welding cable size is selected based on:

- A) The amperage and duty cycle
- B) The type of electrode being used
- C) The length of the cable only
- D) The colour of the insulation

Q29. What is "burn-back" in GMAW?

- A) Porosity from shielding gas loss
- B) The wire burning back to the contact tip
- C) Undercutting at high wire feed speeds
- D) Excessive spatter on the base metal

Q30. What is the difference between a groove weld and a fillet weld?

- A) Groove welds are partial penetration only; fillet welds are always full penetration
- B) Groove welds fill a prepared groove; fillet welds join surfaces at an angle
- C) Groove welds are used for lap joints; fillet welds are used for butt joints
- D) Groove welds require GTAW; fillet welds can use SMAW only

Q31. What is the purpose of a welding screen or curtain?

- A) To reduce heat loss from the weld
- B) To contain welding fumes
- C) To reduce noise from the arc
- D) To protect others from arc radiation

Q32. What is the main characteristic of an E6010 electrode?

- A) Rutile coating, easy slag removal, AC or DC
- B) High cellulosic coating, deep penetration, DCEP only
- C) Iron powder coating, high deposition rate
- D) Low hydrogen, requires dry storage

Q33. Why is AC polarity used for GTAW of aluminum?

- A) AC lowers total heat input so thin aluminum will not burn through
- B) AC gives deeper penetration into aluminum than straight polarity DCEN
- C) AC prevents tungsten contamination
- D) The EP half-cycle removes the aluminum oxide layer

Q34. Oxygen and acetylene cylinders must be stored how far apart when not in use?

- A) At least 3 m (10 ft)
- B) At least 6 m (20 ft)
- C) At least 1.5 m (5 ft)
- D) They may be stored together if capped

Q35. What is the effect of increasing arc length in SMAW beyond the optimal distance?

- A) Reduces heat input
- B) Increases penetration
- C) Causes porosity and spatter
- D) Improves fusion at the weld toes

Q36. What is the neutral flame setting in oxyfuel welding?

- A) Equal volumes of oxygen and acetylene
- B) Oxygen valve fully open, acetylene barely open
- C) Excess oxygen, no feather
- D) Excess acetylene, large feather present

Q37. What is the leg of a fillet weld?

- A) The height of the weld reinforcement above the base metal
- B) The distance from the weld root to the toe
- C) The total length of the fillet weld deposit
- D) The total depth of penetration into the base metal

Q38. What is the maximum working (delivery) pressure allowed for acetylene at the torch?

- A) 207 kPa (30 psi)
- B) 345 kPa (50 psi)
- C) 103 kPa (15 psi)
- D) 690 kPa (100 psi)

Q39. Which shielding gas mixture is most commonly used for GMAW of mild steel?

- A) 50% Argon / 50% Helium
- B) 100% carbon dioxide
- C) 100% Argon
- D) 75% Argon / 25% CO₂

Q40. What is hydrogen-induced cracking (HIC) and which electrode type is most susceptible?

- A) Cracking from rapid cooling and quenching of the weld; low-hydrogen E7018 electrodes most susceptible
- B) Cracking from slag entrapment between passes; iron powder E7024 electrodes most susceptible
- C) Cracking from excessive amperage; rutile E6013 electrodes most susceptible
- D) Cracking from dissolved hydrogen in the HAZ; high-cellulosic electrodes most susceptible

Q41. What is the gas lens in a GTAW torch?

- A) A gas pressure regulator built into the torch body
- B) A glass lens that focuses the shielding gas into a beam
- C) A lens that focuses the arc onto a smaller area for precision
- D) A mesh screen that produces smooth, laminar gas flow

Q42. What is the approximate spray transition current for ER70S-6 wire (0.9 mm / 0.035") with 98% argon / 2% oxygen shielding gas?

- A) Approximately 230–250 A
- B) Approximately 80–100 A
- C) Approximately 165 A
- D) Spray transfer is possible with any gas above 200 A

Q43. What polarity is typically used for FCAW-G?

- A) AC (alternating current)
- B) DCEN (electrode negative)
- C) DCEP (electrode positive)
- D) DCEN for indoor, DCEP for outdoor

Q44. What causes a "crater crack" at the end of a SMAW weld?

- A) Rapid contraction of the pool at arc break
- B) Using DCEN instead of DCEP
- C) Moisture in the low-hydrogen electrode coating
- D) Excessive amperage throughout the weld

Q45. A visual inspection finds slight undercut along the toe of a completed fillet weld. What decides whether that discontinuity is recorded as a rejectable defect?

- A) Whether it can be seen without a magnifying lens or mirror
- B) Any amount of undercut is automatically a rejectable defect
- C) Whether it exceeds the acceptance criteria of the governing code
- D) Whether the welder can grind it out and reweld before handover

Q46. What is the purpose of a Charpy V-notch impact test?

- A) To measure the toughness of the weld at a set temperature
- B) To measure the hardness of the HAZ
- C) To measure the tensile strength of the weld metal
- D) To measure porosity in the weld cross-section

Q47. What colour coding identifies a pure tungsten electrode (EWP)?

- A) Green
- B) Gold
- C) Red
- D) Grey

Q48. What does CTWD stand for in GMAW?

- A) Contact Tip Wear Detection
- B) Contact Tip to Work Distance
- C) Circuit Travel Wire Direction
- D) Current to Wire Diameter ratio

Q49. What type of slag system does E71T-1 flux-cored wire use?

- A) Iron powder — high deposition rate for flat/horizontal only
- B) Cellulosic slag — deep penetrating, used outdoors
- C) Basic slag — used for positional welding with DCEN
- D) Rutile slag — fluid, easily removed, all-position

Q50. A welder is welding in a low-lying area and notices dizziness. The most likely hazard is:

- A) Nitrogen narcosis from shielding gas
- B) Carbon monoxide from the electrode coating
- C) UV radiation causing nausea
- D) Oxygen displacement by argon or CO₂

Answer Key & Explanations

Q1. Answer: D

Anti-spatter spray is applied to nozzles and workpiece surfaces to prevent spatter from adhering and bonding. It reduces nozzle cleaning frequency and cleanup time.

Q2. Answer: C

Atmospheric testing with a calibrated gas monitor must occur before entry to detect flammable gas, oxygen deficiency, or toxic vapours. Ventilation, an attendant posted outside and personal protective equipment are all required as well, but the test result is what tells you whether entry — and hot work inside — is permitted at all, so the test comes first.

Q3. Answer: B

A stringer bead is a narrow, straight pass with no side-to-side weaving motion. It provides better mechanical properties (especially toughness) than a wide weave bead.

Q4. Answer: B

For DCEN GTAW (steel, stainless, titanium), the tungsten is ground to a tapered point. For AC GTAW (aluminum), a rounded ball forms naturally at the tip.

Q5. Answer: C

P91 (9% Cr, 1% Mo) creep-resistant steel requires matching filler metal ER90S-B9 (or equivalent). Standard carbon steel or stainless filler would produce mismatched mechanical properties.

Q6. Answer: B

HF start initiates the arc via a high-frequency spark across the gap between the electrode and workpiece — no contact required. This prevents tungsten contamination and electrode tip damage.

Q7. Answer: D

Rank the modes by arc length and you rank them by voltage. Short circuit sits at the bottom: the arc is so short that the wire dips into the puddle and extinguishes the arc many times a second. Globular sits in the middle, with a longer arc carrying drops larger than the wire diameter across it. True axial spray sits at the top, because the arc has to be long enough and hot enough to pinch the wire tip into a fine stream of droplets smaller than the wire. Pulsed spray is the trap here: its peak current does climb above the spray transition, but the whole point of pulsing is to get spray-type droplet transfer at a lower average voltage and current, which is exactly what lets it run out of position. Voltage alone is not enough either — true spray also needs an argon-rich shielding gas, and a CO₂-rich gas will not give axial spray at any voltage setting.

Q8. Answer: C

Preheat slows the cooling rate of the HAZ, reducing hardness and the risk of hydrogen-induced cracking, especially in high-carbon or low-alloy steels.

Q9. Answer: C

Oxy-fuel cutting works by rapid oxidation of iron, so it is confined to carbon and low-alloy steel. It cannot cut stainless steel — stainless is ferrous, but its chromium forms a refractory oxide that will not burn away — and it cannot cut aluminum or the other non-ferrous metals. Plasma arc cutting melts the metal with a high-temperature ionized gas jet and blows it out of the kerf, so it handles stainless, aluminum and copper readily. Plasma does need an electrically conductive workpiece with the work lead clamped to it, so it offers nothing on materials that do not conduct. Galvanized steel cuts perfectly well with oxy-fuel, and on heavy carbon steel plate oxy-fuel remains the practical choice, since plasma capacity falls away as thickness climbs.

Q10. Answer: C

Low arc voltage relative to wire feed speed causes an unstable arc and excessive spatter in FCAW-G. Proper voltage-WFS balance is critical. Also, 100% CO₂ produces more spatter than Ar/CO₂ mixes.

Q11. Answer: C

Silicon (Si) and manganese (Mn) are deoxidizers. ER70S-6's higher Si/Mn content scavenges oxygen from the weld pool, improving weld quality on slightly rusty or dirty base metal and giving better wetting.

Q12. Answer: D

Read E71T-1 one character at a time. E is electrode. 7 is the minimum tensile strength, 70 000 psi. Do not convert that number yourself: AWS publishes a separate metric classification, and in it the same wire is E491T-1, where 49 stands for 490 MPa. AWS A5.20/A5.20M:2005 Table 1 requires 70–95 ksi for the E7XT-1 classifications and Table 1M requires 490–670 MPa for E49XT-1, and the specification states that the two systems 'are not equivalent, each system must be used independently of the other.' The 1 that sits ahead of the T is the position digit and means all-position. T means the wire is tubular, that is flux-cored, as opposed to a solid GMAW wire. The -1 that follows the T is the usability designator: it identifies a rutile-type, fast-freezing slag system run on DCEP with CO₂ or an argon/CO₂ mix as the shielding gas. The two 1s in the designation therefore carry different meanings, and it is the trailing one that describes gas and polarity. Also keep the shielding split straight: E71T-1 requires an external shielding gas (FCAW-G), while a self-shielded wire such as E71T-11 or E71T-8 generates its own shielding from the core (FCAW-S) and runs with no gas bottle at all.

Q13. Answer: C

Acetylene has an LEL of 2.5% and an UEL of 81% in air, making it one of the most flammable industrial gases.

Q14. Answer: C

The HAZ is the region of base metal adjacent to the fusion zone that has not melted but experienced temperatures high enough to alter its microstructure and properties (grain growth, hardening).

Q15. Answer: A

Low-hydrogen electrodes absorb moisture rapidly. Storage in a rod oven at 120–150°C keeps the coating dry, and a dry coating is what keeps diffusible hydrogen out of the weld metal — moisture picked up by the flux is the source of the hydrogen behind hydrogen-induced (cold) cracking and porosity.

Q16. Answer: B

The feed path controls how smoothly the wire reaches the arc. Worn or wrongly grooved drive rolls, incorrect drive tension, a kinked or clogged liner, a worn contact tip or a tangled spool make the wire arrive in surges, so the arc lengthens and shortens, pops, and burns back. The other faults leave the feed alone and show themselves differently: low shielding gas flow shows up as porosity, too slow a travel speed as excessive buildup and cold lap, and a wrong work angle as undercut and poor bead placement. Check the feed system first — liner condition and length, drive roll size and groove type for the wire being run, drive tension, and the contact tip.

Q17. Answer: C

$CE = C + Mn/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15$. Higher CE means greater hardenability of the base metal, requiring preheat to slow cooling and reduce HIC risk.

Q18. Answer: A

Globular transfer occurs at intermediate voltages between short circuit and spray thresholds. Large irregular droplets — bigger than the wire diameter — form and fall into the pool, causing significant spatter. Generally avoided in production.

Q19. Answer: C

These are worm tracks, also called gas tracks: shallow tunnels or grooves in the surface of the bead, hidden under the slag until it comes off. The flux core generates gas as it burns, and most of that gas escapes up through the puddle. When the slag blanket freezes over the top before the last of it is out, the gas is trapped at the interface and leaves its path printed along the crown. That makes worm tracks a surface signature, quite unlike the fine scattered voids through the body of the bead that come from a damp consumable. Too long an arc is the usual reason — it gives the core more gas to shed and less time in which to shed it — and wire that has picked up moisture adds to the gas load, so shorten the arc, slow down enough to let the puddle degas, and use wire that has been stored dry. The other three answers name real flux-cored defects with different signatures: undercut is a groove in the base metal at the toe, a slag inclusion is non-metallic material buried inside the deposit, and an arc strike is a mark outside the weld itself.

Q20. Answer: A

OFC works by preheating steel to kindling temperature (~870°C/1600°F), then directing a high-pressure oxygen jet at the preheated metal to oxidize (burn) it rapidly.

Q21. Answer: C

Oxygen under high pressure can cause oil and grease to ignite spontaneously (oxygen-enriched combustion). This can cause violent fires or explosions.

Q22. Answer: C

In Canada welding ventilation is an exposure-limit duty, not a fixed airflow number. Provincial regulation makes CSA W117.2 (Safety in welding, cutting, and allied processes) mandatory and then requires capture at source: British Columbia's OHS Regulation section 12.112 requires welding, cutting and similar processes to be carried out according to CSA W117.2, and section 12.114 requires effective local exhaust ventilation at any fixed work station to minimize worker exposure to harmful air contaminants. The test of adequacy is whether fume and gas stay below the occupational exposure limits for the metals and coatings involved — use the edition of W117.2 and the exposure limits your province adopts. The 2,000 cfm per welder figure offered here is American, not Canadian: OSHA 1910.252 sets that rate where the space is less than 10,000 cubic feet per welder or the ceiling is less than 16 feet, and it has no force here. The 1,000 cfm variant simply invents a Canadian version of that same US ceiling-height trigger. The idea that ventilation only matters above some current is false — fume is produced at every setting, and there is no amperage below which ventilation stops being required. The one Canadian airflow figure worth memorizing comes from the Canadian Centre for Occupational Health and Safety: a moveable exhaust hood should provide an air velocity of at least 100 ft/min (0.5 m/s) across the welding arc, placed as close to the work as practical. Local exhaust is mandatory regardless of general airflow when welding coated, galvanized or stainless material, where zinc oxide and hexavalent chromium exceed exposure limits even in a well-ventilated shop. AWS D1.1 and CSA W59 are fabrication codes and set no ventilation requirement.

Q23. Answer: C

DCEP (reverse polarity) means the electrode is connected to the positive terminal and the work to the negative terminal. In SMAW this is the deeper-penetrating polarity, and it is what E6010 and E7018 are run on. Do not carry that rule across to GTAW: the tungsten there is not consumed, and DCEN is the polarity that concentrates the heat in the work and gives the deeper penetration. The rule about which polarity penetrates belongs to the process, not to the letters.

Q24. Answer: A

"Walking the cup" means bracing the ceramic nozzle against the pipe and using a rocking or rolling motion to move the torch. This provides a stable torch angle and consistent arc length for pipe welding.

Q25. Answer: C

The liner (conduit liner) guides the wire electrode from the drive rolls through the torch body to the contact tip, preventing kinking and ensuring smooth wire feed.

Q26. Answer: C

E6013 (rutile-coated) produces a soft arc, light spatter, easy slag removal, and shallow penetration — making it ideal for thin sheet metal where burn-through is the primary risk. Wrong answers: E6010 and E6011 have deep penetration and forceful arc (designed for root passes and pipe), which would easily burn through thin material. E7018 (low-hydrogen) is excellent for structural work but requires an oven and produces a broader bead; it is not the preferred choice for light gauge sheet. E6013's easy restriking and smooth bead also make it a good learning electrode, but its low penetration means it's NOT suitable for code-quality root passes or high-strength joints where full fusion is required.

Q27. Answer: A

"Sugaring" is a rough, oxidized surface on the weld root caused by atmospheric oxygen reacting with chromium in the hot metal, forming chromium oxides (Cr_2O_3) and depleting the corrosion-resistant chromium from the surface.

Q28. Answer: A

Cable must handle the amperage and duty cycle of the welding process without overheating. Undersized cable creates excessive resistance, heat, and fire hazard.

Q29. Answer: B

Burn-back occurs when the wire electrode burns back and fuses to the contact tip, stopping the weld. Caused by low wire feed speed relative to voltage.

Q30. Answer: B

A groove weld occupies a groove or gap prepared in the base metal (butt joint). A fillet weld is a triangular cross-section weld joining surfaces at approximately 90° (T-joint, lap joint, corner joint).

Q31. Answer: D

Welding screens protect nearby workers and bystanders from arc flash (UV/IR radiation), which can cause arc eye (photokeratitis) and skin burns.

Q32. Answer: B

E6010 has a high-cellulosic sodium coating that produces a forceful, deep-penetrating arc. It requires DCEP and is ideal for root passes on pipe and dirty/rusty steel.

Q33. Answer: D

Aluminum has a tenacious oxide layer (Al_2O_3 , melting point 2050°C) that prevents fusion. During the electrode positive (DCEP) half of AC, electrons flow from the work to the electrode, breaking up the oxide — this is cathodic cleaning.

Q34. Answer: B

CSA regulations require oxygen and fuel gas cylinders to be stored at least 6 m apart or separated by a 1.5 m high fire-resistant wall.

Q35. Answer: C

Longer arc length increases arc voltage and heat, but also exposes the molten weld pool to atmospheric contamination, causing porosity and spatter. Proper arc length equals electrode diameter.

Q36. Answer: A

A neutral flame has a 1:1 oxygen-to-acetylene ratio, producing a sharp bright inner cone with no feather — suitable for most welding.

Q37. Answer: B

The leg of a fillet weld is the distance from the root of the joint to the toe of the weld, measured along the fusion face. For an equal-leg fillet, both legs are the same length.

Q38. Answer: C

Free acetylene gas is chemically unstable and can decompose explosively on its own, without any oxygen present, once its pressure climbs too high — so the fuel regulator is never set above 103 kPa (15 psi) gauge. Note what this limit applies to: it is the delivery pressure the regulator feeds to the torch, not the cylinder pressure. A full acetylene cylinder reads far higher, roughly 1700 kPa (250 psi), and that is safe only because the acetylene inside is dissolved in acetone held in a porous filler rather than existing as free gas. For the same reason the withdrawal rate is limited, commonly to about one-seventh of cylinder capacity per hour, so that acetone is not dragged out with the gas.

Q39. Answer: D

C25 (75% Ar / 25% CO₂) is the most widely used mixture for mild steel GMAW. It provides a stable arc, good penetration, and lower spatter than 100% CO₂.

Q40. Answer: D

HIC (cold cracking) occurs when hydrogen from the electrode diffuses into the HAZ and combines with residual stress and microstructure to cause delayed cracking. Cellulosic (E6010/E6011) electrodes produce the most hydrogen.

Q41. Answer: D

A gas lens replaces the standard collet body. Its porous metal mesh screen produces a smooth, laminar (non-turbulent) flow of shielding gas, extending coverage and enabling longer electrode stickout.

Q42. Answer: C

Spray transfer requires an argon-rich gas (at least ~80% Ar). With 98Ar/2O₂, 0.9 mm ER70S-6 transitions from globular to spray at roughly 165 A; with 90Ar/10CO₂ the transition rises to about 200 A. With C25 (75/25) true axial spray cannot be achieved — the high CO₂ content disrupts the arc column.

Q43. Answer: C

Gas-shielded flux-cored wire is run on DCEP, direct current electrode positive, the same polarity as GMAW solid wire. AWS A5.20/A5.20M:2005 Table 2, Electrode Usability Requirements, defines the term in a footnote as direct current electrode positive, or reverse polarity, and its polarity column is where the word "typically" in the question comes from. The gas-shielded usability designators are 1, 2, 5, 9 and 12. Designators 1, 2, 9 and 12, which include the widely used E71T-1, are listed for DCEP, and the annex describes the T-1 wires as designed for single and multiple pass welding using DCEP. Designator 5 is the exception: the flat-and-horizontal E70T-5C and E70T-5M electrodes are listed DCEP, but the all-position E71T-5C and E71T-5M electrodes are listed "DCEP or DCEN", with a footnote that some of them are recommended for DCEN to improve out-of-position welding and that the manufacturer should be consulted. Self-shielded wire is where polarity really varies: designators 3, 4 and 6 are DCEP, while 7, 8, 10, 11, 13 and 14, the majority and the common E71T-8 and E71T-11 among them, are DCEN. So a gas-shielded wire is set up electrode positive unless its classification or the manufacturer's data sheet says otherwise, and the leads are never left where the last spool had them: wrong polarity on a flux-cored wire shows up as an erratic arc, heavy spatter, porosity and poor fusion. The polarity column of that table carries only DCEP, DCEN, or "not specified" for the G classifications, so AC is not a classified polarity for any carbon steel flux-cored designator. Nor does any designator tie polarity to where the work is done: indoors or outdoors decides between gas-shielded and self-shielded wire, not between electrode positive and electrode negative.

Q44. Answer: A

When the arc is suddenly extinguished, the weld pool shrinks rapidly. If the crater is not filled, tensile stresses from contraction cause a star-shaped crater crack.

Q45. Answer: C

Almost every weld contains some discontinuity. A discontinuity only becomes a defect when it falls outside the acceptance criteria of the code or job specification the work is being performed to, so the same undercut can be acceptable on one job and rejectable on another. The Red Seal Occupational Standard makes this the welder's call to determine acceptability according to job specifications and codes, and to keep records of what was found. Repairability does not change the accept or reject decision, and inspection aids such as magnifying lenses and mirrors are normal visual-inspection tools, not the dividing line.

Q46. Answer: A

The Charpy test strikes a notched specimen with a pendulum and measures the energy absorbed before fracture (in joules). It evaluates toughness at specified temperatures, critical for pressure vessels and low-temperature service.

Q47. Answer: A

AWS A5.12M/A5.12 (a modified adoption of ISO 6848) colour coding: pure tungsten, classification EWP, is Green. EWP is unalloyed tungsten, 99.5% minimum, with no oxide addition. It is used with alternating current for welding aluminum and magnesium, where it holds a clean balled end. The other three colours in this question belong to different electrodes: Grey is 2% ceriated (EWCe-2), Red is 2% thoriated (EWTh-2), and Gold is 1.5% lanthanated (EWLa-1.5). One change catches people out on the exam and in the rod drawer: the 2009 edition of A5.12 moved EWCe-2 from orange to grey, and grey is no longer the EWG code. Older charts and older stock still show orange for ceriated, so trust the current colour band, not the wall chart.

Q48. Answer: B

CTWD (Contact Tip to Work Distance) is the distance from the contact tip to the workpiece. It affects electrical stickout and therefore current/deposition rate.

Q49. Answer: D

E71T-1 is a rutile-type flux-cored wire used with CO₂ or Ar/CO₂ shielding gas. It produces a fluid, easily removed slag and is suitable for all-position welding.

Q50. Answer: D

Argon and CO₂ are heavier than air and can accumulate at floor level in low-lying areas, displacing oxygen and causing asphyxiation.

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